LESSON PLAN

PISTON ENGINE (15)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	02
Total No. of Lesson Plans	6	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	IC Engines, Variations and usage in aviation industry as main propulsion plant and auxiliary machinery, operation cycles of four stroke and
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	IC Engines, Variations and usage in aviation industry as main propulsion plant and auxiliary machinery, operation cycles of four stroke and two stroke Piston engine, comparison, advantages, modified Otto cycle
5. Specific Objectives	15.1.1.2 Trainees will be able to explain Otto cycle and operation of four stroke piston engine.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of PPT diagrams; Instructor supervision of diagram tracing; Safety: No physical engine access during session.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of an internal combustion engine?
(b) Name two common types of piston engines used in aviation.
(c) Define the term 'Otto cycle'.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Basic principles of IC engines • Four stroke cycle operation • Modified Otto cycle explanation	L & Ds	PPT, IB	15 min
(b)	• Two stroke vs Four stroke comparison • Aviation usage as main propulsion • Auxiliary machinery applications	L & Ds	PPT, PDF	20 min
(c)	• Advantages of four stroke engines • Engine variations in aviation • Summary of operation cycles	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the four strokes of the Otto cycle in order.
(b) Compare the advantages of a four stroke engine over a two stroke engine.

12. Home Assignment and Follow Up	Prepare a brief note on how the Otto cycle is modified for aviation piston engines, referencing the next lesson on superchargers.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Principles & operations of Super chargers, description and operations.	Refer to the master syllabus reference list	LP 03

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min